# MSG No. 90.99 $P4'2'_12$ [ Type III, tetragonal ]

Table 1: Wyckoff site: `2a`, site symmetry: `2.22`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ 0]$ | `[1,2,3,4]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | `[5,6,7,8]` |

Table 2: Wyckoff site: `2b`, site symmetry: `2.22`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ 0,\ \frac{1}{2}\right]$ | `[1,2,3,4]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[5,6,7,8]` |

Table 3: Wyckoff site: `2c`, site symmetry: `4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ z\right]$ | `[1,2,5,6]` |
| 2 | $\left[\frac{1}{2},\ 0,\ -z\right]$ | `[3,4,7,8]` |

Table 4: Wyckoff site: `4d`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,2]` |
| 2 | $[0,\ 0,\ -z]$ | `[3,4]` |
| 3 | $\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$ | `[5,6]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$ | `[7,8]` |

Table 5: Wyckoff site: `4e`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ x,\ 0]$ | `[1,3]` |
| 2 | $[-x,\ -x,\ 0]$ | `[2,4]` |
| 3 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0\right]$ | `[5,8]` |
| 4 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$ | `[6,7]` |

Table 6: Wyckoff site: `4f`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ x,\ \frac{1}{2}\right]$ | `[1,3]` |
| 2 | $\left[-x,\ -x,\ \frac{1}{2}\right]$ | `[2,4]` |
| 3 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | `[5,8]` |
| 4 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | `[6,7]` |

Table 7: Wyckoff site: 8g, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, y,\, z]$ | [1] |
| 2 | $[-x,\, -y,\, z]$ | [2] |
| 3 | $[y,\, x,\, -z]$ | [3] |
| 4 | $[-y,\, -x,\, -z]$ | [4] |
| 5 | $\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z\right]$ | [5] |
| 6 | $\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z\right]$ | [6] |
| 7 | $\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$ | [7] |
| 8 | $\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$ | [8] |